4 Multiple Integrals

4.2.1 Changing the Order of Integration

Example 4.2.1 Evaluate the integral

$$\int_0^1 \int_x^1 \sin(y^2) dy dx.$$

Example 4.2.2 Evaluate the integral

$$\int_0^1 \int_{3y}^3 e^{x^2}\, dx dy.$$

Theorem 4.2.3 *(Properties of Double Integrals)*

1. $\iint_D (f_1(x,y) + f_2(x,y))\, dA = \iint_D f_1(x,y)\, dA + \iint_D f_2(x,y)\, DA.$
2. $\iint_D cf(x,y)\, dA = c \iint_D f(x,y)\, dA.$
3. $\iint_D f(x,y)\, dA = \iint_{D_1} f(x,y)\, dA + \iint_{D_2} f(x,y)\, dA$, *where* $D = D_1 \cup D_2$ *and* D_1 *and* D_2 *do not overlap, except possibly on their boundaries.*
4. $\iint_D 1\, dA = A(D).$
5. *If* $m \leq f(x,y) \leq M$ *on* D, *then* $mA(D) \leq \iint_D f(x,y)\, dA \leq MA(D).$

4.3 Double Integrals in Polar Coordinates

Theorem 4.3.1 *If f is a continuous function on a polar rectangle R given by $a \leq \theta \leq b$ and $\alpha \leq r \leq \beta$, then*

$$\iint_R f(x,y)\,dA = \int_\alpha^\beta \int_a^b r f(r\cos\theta, r\sin\theta)\,d\theta dr.$$

Example 4.3.2 Evaluate the integral

$$\iint_R (3x + 4y^2)\,dA$$

where R is the region in the upper-half plane bounded by the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

Example 4.3.3 Find the volume of the solid bounded by the plane $z = 0$ and the surface $z = 1 - x^2 - y^2$.

Example 4.3.4 Use a double integral to find the area of the region enclosed by one loop of the function $r = \cos 2\theta$.

Example 4.3.5 FInd the volume of the solid that lies in the cylinder $x^2 + y^2 = 2x$ and between the xy-plane and the surface $z = x^2 + y^2$.